Predicting Persistent Overtime among Fixed-Term Staff of the Metropolitan Health Services

C2 · Train and Explain our First Model (Classification)

Group 01 · Alessandro Lavezzi · José Saavedra
Instructor: Tomás Fontecilla Correa · Santiago, November 8, 2026
Notebooks: *01_W2_project.ipynb* (preparation and split) and *02_C2_classification.ipynb* (model)

1. Question and task

Since W1 the project has measured **persistent overtime**: a person is persistent when they receive overtime in at least 75% of their payroll months. C2 moves from description to prediction. **On September 1, 2025, using only what was known that day, can we anticipate who will be persistent in the following 12 months (September 2025 – August 2026)?** The decision it supports is which people, and the positions behind them, to review first for staffing.

The source is the monthly payroll of fixed-term (*contrata*) staff of the six Santiago Metropolitan Health Services, published under Active Transparency (Law 20.285). One row is one person, identified by service + name, because the files contain no national ID. The target is persistent in year 2 (1) or not (0), and the **positive class is persistent** (18.3% of people). The inputs are known at decision time and come from the **feature window, September 2024 – July 2025**: months on the payroll, percentage of months with overtime, mean overtime hours, staff category and overtime entitlement in the last month, and the service.

2. Response to the W2 feedback

Table 1
Instructor feedback on W2 and changes in C2

Feedback	What we found	Change in C2
Build cohorts and variables according to the decision date	The draft used the August 2025 payroll, which is published about one month later, and selected people with a rule that needs the future (≥ 6 months in year 2)	Cohort: ≥ 6 months in the feature window and on the July 2025 payroll (8,732 people). August 2025 is a gap month. 598 people without a reliable year-2 label are reported, not hidden
Investigate name matches across services	135 names appear in two services: 39 in the same month (homonyms or two jobs) and 96 one after another (likely transfers)	The 24 C2 people concerned (2 of them in different split sets) were excluded from all sets
Investigate lags between month worked and month paid	372 contract rows with overtime (0.66%) are paid after the contract ended: the payroll month is the payment month	The gap month limits the lag; it remains a limitation

Note. Retained from W2: the person-level split, the 75% threshold and the rule "no variable from the period we predict". Notebook 02, section 1.0.

3. Method

The 8,110 labelled people keep the set assigned in W2: a split by person, stratified by service and staff category, with seed 42. This gives **5,691 people for training** (1,043 persistent), **1,227 for validation** (238) and **1,192 for test** (204). Because the same person is never in two sets, the model cannot memorize individuals.

- **Reference rule (baseline):** "persistent in the feature window → persistent in year 2". It has nothing to fit, and it is the simplest forecast a manager could make with the same data.

- **Model: logistic regression.** It is simple, and its coefficients can be read and discussed. Numeric inputs are standardized and categories are one-hot encoded; it uses the default L2 penalty (C = 1) without tuning. Scaling, encoding and coefficients are fitted in one pipeline on the **training set only**.
- **Threshold:** chosen on **validation** with a rule fixed beforehand: from 0.10 to 0.90 in steps of 0.05, highest F1, ties broken by higher recall. The result is **0.35**; with an 18% positive class, predicted probabilities rarely exceed 0.5.
- **Test set:** used once, after the model and the threshold were frozen.

4. Results

Table 2

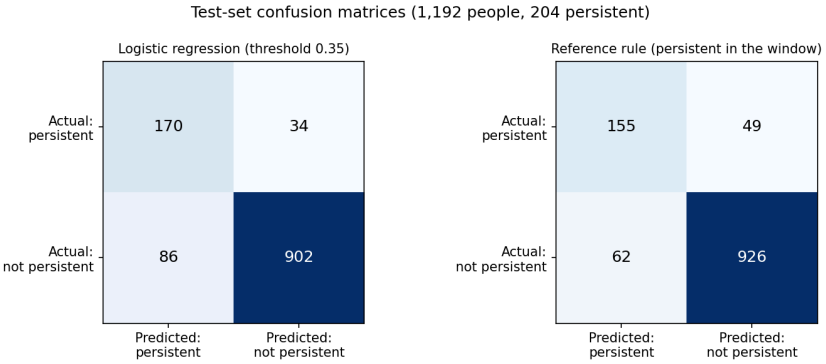
Model and reference rule evaluated on the same people (positive class = persistent)

Model or rule	Set	TP	FP	FN	TN	Precision	Recall	F1	Accuracy
Reference rule	Validation	171	61	67	928	0.737	0.718	0.728	0.896
Logistic regression (0.35)	Validation	197	71	41	918	0.735	0.828	0.779	0.909
Reference rule	**Test**	155	62	49	926	0.714	0.760	0.736	0.907
Logistic regression (0.35)	**Test**	170	86	34	902	0.664	0.833	0.739	0.899
Always "not persistent"	Test	0	0	204	988	0.000	0.000	0.000	0.829

Note. Test: 1,192 people, 204 persistent. Notebook 02, sections 3.3 and 4.1.

Figure 1

Final test-set confusion matrices (rows = actual, columns = predicted)



Note. Notebook 02, section 4.1.

On the test set the model finds **170 of the 204** persistent people (recall 0.833), against 155 for the reference rule (0.760), but it raises **24 more false alarms** (86 vs 62). The two effects cancel out in F1: **0.739 vs 0.736**. A **false positive** triggers a staffing review where overtime would not persist, costing review time and possibly an unnecessary hire. A **false negative** leaves a recurring case unreviewed for another year of overtime, paid with a 25–50% surcharge. Accuracy alone is misleading: predicting that nobody is persistent reaches 0.829 while finding no one, and the rule even beats the model in accuracy (0.907 vs 0.899) because it flags fewer people. In the coefficients, past persistence dominates: one standard deviation more of the percentage of months with overtime multiplies the odds of being persistent by about 6 (odds ratio 5.9). Holding it fixed, overtime entitlement raises the odds (OR 1.9), while professionals, technicians (OR ≈ 0.63) and the Sur service (OR 0.46) have lower odds. These are associations, not causes.

5. Interpretation, limits and next step

What the model adds. The model does not clearly beat the rule in F1. Its advantage on validation (0.779 vs 0.728) almost disappeared on test, and with about 200 positives per set a difference of that size is within sampling variation. What it adds is a different trade-off, finding more persistent people at the cost of more false alarms. It also gives a **probability for each person**, so management can rank cases and set the threshold to match how many reviews it can carry out.

Most important limit. There is only **one decision date and one predicted year**, so we cannot show that the model works in another year, for example after changes such as Dictamen E121666. Other limits: people are identified by name only; payments are recorded in the month paid, not the month worked; the 598 people who left cannot be evaluated; and the test set had already been seen descriptively in W2.

Possible improvement. Choose the threshold with **explicit costs** instead of F1: the payroll gives the overtime amount that a missed case costs, which can be compared with the cost of a review. When the 2026–27 payroll is published, add a second cohort and validate across years. Finally, a useful prediction does not prove that an action works: showing that reviews or hires reduce overtime needs a comparison group, for example a pilot in some flagged units.

References

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